Praveenkumar G

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Date of birth: 08 Aug 2005

PROFESSIONAL INTRODUCTION

Technology professional with an academic background in Electronics and Communication Engineering, including digital electronics, control systems, microprocessors, and programming. Currently working as a DevSecOps Industrial Trainee, with knowledge of AWS, CI/CD, infrastructure as code, containers, and orchestration tools. Interested in cloud computing, DevOps, DevSecOps, cybersecurity, and applied AI/ML. Brings problem-solving and analytical thinking skills developed through academic and personal projects. Enjoys building practical solutions, exploring emerging technologies, and continuously learning.

PROFESSIONAL EXPERIENCE

DevSecOps Industrial Trainee | **Baton Systems** March 2, 2026 – Present

Taramani, Chennai

TECHNICAL SKILLS

Programming and Scripting: Python, C, Verilog (HDL), HTML, CSS

Cloud Computing (AWS): AWS

DevOps and CI/CD: Jenkins, Argo CD, Ansible

Infrastructure as Code: Terraform, Atlantis

Containers and Orchestration: Docker, Kubernetes

Operating Systems and Networking: Linux, Operating Systems, Computer Networks, Wireshark

AI/ML Libraries and Frameworks: TensorFlow, OpenCV, scikit-learn, NumPy, Pandas

Other Relevant Tools and Concepts: Git, GitHub, VS Code, SQL, DBMS, OOP, DSA, Digital Electronics, Control Systems, Microprocessors, Raspberry Pi, Arduino UNO

EDUCATION

Madras Institute of Technology, Anna University | **B.E. in Electronics and Communication Engineering** 2022 – 2026

CGPA: 8.82/10.0

SRM Muthamizhl Hr. Sec. School, Salem 2020 – 2022

HSC: 97.67% | SSLC: 99.67%

PROJECTS

Orvion October 2026

Personal workspace application

Developed a personal workspace for everyday plans, records, and activity. After sign-in, users can navigate a dashboard and sections for tasks, notes, calendar, finance, pending expenses, entertainment, timeline, analytics, and settings.

Rice Grain Quality Detection January 2026

Computer vision, YOLOv8, CNNs, Streamlit

Developed an automated system to detect rice grains, classify varieties and defects, and assign a quality grade based on the percentage of defective grains. Combined YOLOv8 object detection with custom CNNs and integrated the models into a Streamlit application for assessment of bulk rice images.

Multi-trait Fingerprint Analysis October 2025

Deep learning, CNNs, Streamlit

Built a non-invasive deep-learning framework to predict blood group, gender, and finger type from fingerprint images. Used two custom 22-layer CNNs to learn ridge patterns and minutiae-level features, with a Streamlit application for real-time prediction from an uploaded image for research and educational use.

Smart Agricultural Monitoring System April 2025

Raspberry Pi 4, Python, Twilio API

Built an image-processing and deep-learning system to detect plant diseases; reported 92% accuracy. Deployed on Raspberry Pi 4 and used the Twilio API to send SMS alerts to farmers.

Cyber Threat Prediction February 2025

Python, TensorFlow, Git

Developed a unified deep-learning model to identify DDoS attacks, malware, and system failures from multi-source logs. Achieved 91.7% validation accuracy using dropout and the Adam optimizer with clipnorm.

Library Management System March 2024

Python, SQL, OOP

Created a database application for book and student records, implementing SQL-based create, read, update, and delete operations.

DesiltBot – Silt Removal System March 2024

Raspberry Pi, OpenCV, cloud integration

Developed a Raspberry Pi, camera, and load-cell system to identify, weigh, and remove silt. The project included automated material handling, real-time data logging, and GPS tracking.

Bluetooth-controlled Home Automation February 2023

Arduino UNO, Bluetooth

Created a smart, affordable, and modular system for wirelessly controlling electrical appliances using an Arduino UNO and a Bluetooth-enabled smartphone.

PRESENTATIONS AND PUBLICATIONS

- **IEEE SISIMPACT 2025** — "DDoS Detection in IoT using DL" (November 2025)
- **IJERT** — "Smart Agricultural Monitoring System," ISSN: 2278-0181 (May 2025)
- **National Conference CST'24** — "Reduction of Flicker Noise" (May 2024)

CERTIFICATIONS AND TRAINING

- **NPTEL:** Deep Learning; score: 61%
- **Maven Silicon:** Verilog HDL
- **Great Learning:** Introduction to C; Introduction to Web Development
- **Coursera:** Introduction to Embedded Systems & IoT; Python for Data Analytics; Basics of Python
- **Workshops:** Machine Learning in Python: Learn, Build and Deploy (2024); Engineering Workshop at KLA (2023)

OPEN-SOURCE CONTRIBUTIONS

a11y-shiftleft-cli

- Added an expected-output section to the demo README to help new contributors understand the expected results (PR #28).

CODING PROFILE

GeeksforGeeks

- Solved 154 coding problems across Basic, Easy, and Medium difficulty levels.
- Coding score: 292 | Institute rank: 256.

ACHIEVEMENTS AND LEADERSHIP

- Finalist, HackIT'25, CEG Anna University (2025); Predicathon, Unstop (2024)
- Coordinated placement drives as a Placement Representative and assisted in organizing cultural events.

ADDITIONAL INFORMATION

Languages: Tamil (native), English (full professional proficiency)
Interests: Software development, hardware/software testing, data analysis, DevSecOps
Soft Skills: Problem solving, analytical thinking, teamwork, communication, adaptability, time management